

```

COUNTING-SORT( $A, n, k$ )
1  let  $B[1:n]$  and  $C[0:k]$  be new arrays
2  for  $i = 0$  to  $k$ 
3       $C[i] = 0$ 
4  for  $j = 1$  to  $n$ 
5       $C[A[j]] = C[A[j]] + 1$ 
6  //  $C[i]$  now contains the number of elements equal to  $i$ .
7  for  $i = 1$  to  $k$ 
8       $C[i] = C[i] + C[i - 1]$ 
9  //  $C[i]$  now contains the number of elements less than or equal to  $i$ .
10 // Copy  $A$  to  $B$ , starting from the end of  $A$ .
11 for  $j = n$  downto 1
12      $B[C[A[j]]] = A[j]$ 
13      $C[A[j]] = C[A[j]] - 1$  // to handle duplicate values
14 return  $B$ 

```

Figure 8.2 illustrates counting sort. After the **for** loop of lines 2–3 initializes the array C to all zeros, the **for** loop of lines 4–5 makes a pass over the array A to inspect each input element. Each time it finds an input element whose value is i , it increments $C[i]$. Thus, after line 5, $C[i]$ holds the number of input elements equal to i for each integer $i = 0, 1, \dots, k$. Lines 7–8 determine for each $i = 0, 1, \dots, k$ how many input elements are less than or equal to i by keeping a running sum of the array C .

Finally, the **for** loop of lines 11–13 makes another pass over A , but in reverse, to place each element $A[j]$ into its correct sorted position in the output array B . If all n elements are distinct, then when line 11 is first entered, for each $A[j]$, the value $C[A[j]]$ is the correct final position of $A[j]$ in the output array, since there are $C[A[j]]$ elements less than or equal to $A[j]$. Because the elements might not be distinct, the loop decrements $C[A[j]]$ each time it places a value $A[j]$ into B . Decrementing $C[A[j]]$ causes the previous element in A with a value equal to $A[j]$, if one exists, to go to the position immediately before $A[j]$ in the output array B .

How much time does counting sort require? The **for** loop of lines 2–3 takes $\Theta(k)$ time, the **for** loop of lines 4–5 takes $\Theta(n)$ time, the **for** loop of lines 7–8 takes $\Theta(k)$ time, and the **for** loop of lines 11–13 takes $\Theta(n)$ time. Thus, the overall time is $\Theta(k + n)$. In practice, we usually use counting sort when we have $k = O(n)$, in which case the running time is $\Theta(n)$.

Counting sort can beat the lower bound of $\Omega(n \lg n)$ proved in Section 8.1 because it is not a comparison sort. In fact, no comparisons between input elements occur anywhere in the code. Instead, counting sort uses the actual values of the

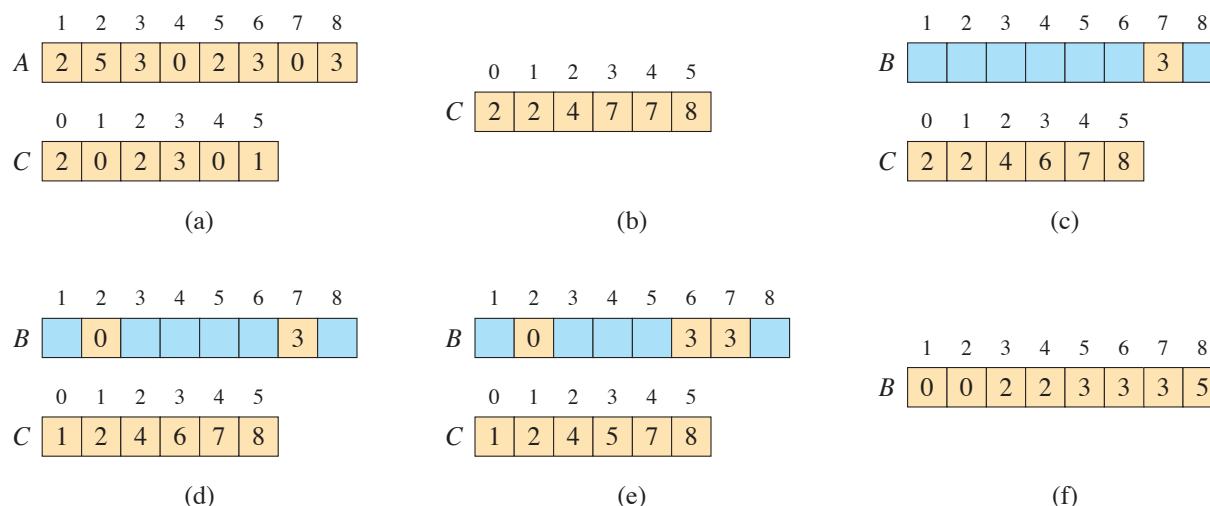


Figure 8.2 The operation of COUNTING-SORT on an input array $A[1:8]$, where each element of A is a nonnegative integer no larger than $k = 5$. (a) The array A and the auxiliary array C after line 5. (b) The array C after line 8. (c)–(e) The output array B and the auxiliary array C after one, two, and three iterations of the loop in lines 11–13, respectively. Only the tan elements of array B have been filled in. (f) The final sorted output array B .

elements to index into an array. The $\Omega(n \lg n)$ lower bound for sorting does not apply when we depart from the comparison sort model.

An important property of counting sort is that it is *stable*: elements with the same value appear in the output array in the same order as they do in the input array. That is, it breaks ties between two elements by the rule that whichever element appears first in the input array appears first in the output array. Normally, the property of stability is important only when satellite data are carried around with the element being sorted. Counting sort's stability is important for another reason: counting sort is often used as a subroutine in radix sort. As we shall see in the next section, in order for radix sort to work correctly, counting sort must be stable.

Exercises

8.2-1

Using Figure 8.2 as a model, illustrate the operation of COUNTING-SORT on the array $A = \langle 6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2 \rangle$.

8.2-2

Prove that COUNTING-SORT is stable.

8.2-3

Suppose that we were to rewrite the **for** loop header in line 11 of the COUNTING-SORT as

11 **for** $j = 1$ **to** n

Show that the algorithm still works properly, but that it is not stable. Then rewrite the pseudocode for counting sort so that elements with the same value are written into the output array in order of increasing index and the algorithm is stable.

8.2-4

Prove the following loop invariant for COUNTING-SORT:

At the start of each iteration of the **for** loop of lines 11–13, the last element in A with value i that has not yet been copied into B belongs in $B[C[i]]$.

8.2-5

Suppose that the array being sorted contains only integers in the range 0 to k and that there are no satellite data to move with those keys. Modify counting sort to use just the arrays A and C , putting the sorted result back into array A instead of into a new array B .

8.2-6

Describe an algorithm that, given n integers in the range 0 to k , preprocesses its input and then answers any query about how many of the n integers fall into a range $[a : b]$ in $O(1)$ time. Your algorithm should use $\Theta(n + k)$ preprocessing time.

8.2-7

Counting sort can also work efficiently if the input values have fractional parts, but the number of digits in the fractional part is small. Suppose that you are given n numbers in the range 0 to k , each with at most d decimal (base 10) digits to the right of the decimal point. Modify counting sort to run in $\Theta(n + 10^d k)$ time.

8.3 Radix sort

Radix sort is the algorithm used by the card-sorting machines you now find only in computer museums. The cards have 80 columns, and in each column a machine can punch a hole in one of 12 places. The sorter can be mechanically “programmed” to examine a given column of each card in a deck and distribute the card into one

329	720	720	329
457	355	329	355
657	436	436	436
839	457	839	457
436	657	355	657
720	329	457	720
355	839	657	839

Figure 8.3 The operation of radix sort on seven 3-digit numbers. The leftmost column is the input. The remaining columns show the numbers after successive sorts on increasingly significant digit positions. Tan shading indicates the digit position sorted on to produce each list from the previous one.

of 12 bins depending on which place has been punched. An operator can then gather the cards bin by bin, so that cards with the first place punched are on top of cards with the second place punched, and so on.

For decimal digits, each column uses only 10 places. (The other two places are reserved for encoding nonnumeric characters.) A d -digit number occupies a field of d columns. Since the card sorter can look at only one column at a time, the problem of sorting n cards on a d -digit number requires a sorting algorithm.

Intuitively, you might sort numbers on their *most significant* (leftmost) digit, sort each of the resulting bins recursively, and then combine the decks in order. Unfortunately, since the cards in 9 of the 10 bins must be put aside to sort each of the bins, this procedure generates many intermediate piles of cards that you would have to keep track of. (See Exercise 8.3-6.)

Radix sort solves the problem of card sorting—counterintuitively—by sorting on the *least significant* digit first. The algorithm then combines the cards into a single deck, with the cards in the 0 bin preceding the cards in the 1 bin preceding the cards in the 2 bin, and so on. Then it sorts the entire deck again on the second-least significant digit and recombines the deck in a like manner. The process continues until the cards have been sorted on all d digits. Remarkably, at that point the cards are fully sorted on the d -digit number. Thus, only d passes through the deck are required to sort. Figure 8.3 shows how radix sort operates on a “deck” of seven 3-digit numbers.

In order for radix sort to work correctly, the digit sorts must be stable. The sort performed by a card sorter is stable, but the operator must be careful not to change the order of the cards as they come out of a bin, even though all the cards in a bin have the same digit in the chosen column.

In a typical computer, which is a sequential random-access machine, we sometimes use radix sort to sort records of information that are keyed by multiple fields. For example, we might wish to sort dates by three keys: year, month, and day. We could run a sorting algorithm with a comparison function that, given two dates,

compares years, and if there is a tie, compares months, and if another tie occurs, compares days. Alternatively, we could sort the information three times with a stable sort: first on day (the “least significant” part), next on month, and finally on year.

The code for radix sort is straightforward. The RADIX-SORT procedure assumes that each element in array $A[1:n]$ has d digits, where digit 1 is the lowest-order digit and digit d is the highest-order digit.

```

RADIX-SORT( $A, n, d$ )
1  for  $i = 1$  to  $d$ 
2      use a stable sort to sort array  $A[1:n]$  on digit  $i$ 

```

Although the pseudocode for RADIX-SORT does not specify which stable sort to use, COUNTING-SORT is commonly used. If you use COUNTING-SORT as the stable sort, you can make RADIX-SORT a little more efficient by revising COUNTING-SORT to take a pointer to the output array as a parameter, having RADIX-SORT preallocate this array, and alternating input and output between the two arrays in successive iterations of the **for** loop in RADIX-SORT.

Lemma 8.3

Given n d -digit numbers in which each digit can take on up to k possible values, RADIX-SORT correctly sorts these numbers in $\Theta(d(n + k))$ time if the stable sort it uses takes $\Theta(n + k)$ time.

Proof The correctness of radix sort follows by induction on the column being sorted (see Exercise 8.3-3). The analysis of the running time depends on the stable sort used as the intermediate sorting algorithm. When each digit lies in the range 0 to $k - 1$ (so that it can take on k possible values), and k is not too large, counting sort is the obvious choice. Each pass over n d -digit numbers then takes $\Theta(n + k)$ time. There are d passes, and so the total time for radix sort is $\Theta(d(n + k))$. ■

When d is constant and $k = O(n)$, we can make radix sort run in linear time. More generally, we have some flexibility in how to break each key into digits.

Lemma 8.4

Given n b -bit numbers and any positive integer $r \leq b$, RADIX-SORT correctly sorts these numbers in $\Theta((b/r)(n + 2^r))$ time if the stable sort it uses takes $\Theta(n + k)$ time for inputs in the range 0 to k .

Proof For a value $r \leq b$, view each key as having $d = \lceil b/r \rceil$ digits of r bits each. Each digit is an integer in the range 0 to $2^r - 1$, so that we can use counting sort with $k = 2^r - 1$. (For example, we can view a 32-bit word as having four 8-bit digits, so that $b = 32$, $r = 8$, $k = 2^8 - 1 = 255$, and $d = b/r = 4$.) Each pass of counting sort takes $\Theta(n + k) = \Theta(n + 2^r)$ time and there are d passes, for a total running time of $\Theta(d(n + 2^r)) = \Theta((b/r)(n + 2^r))$. ■

Given n and b , what value of $r \leq b$ minimizes the expression $(b/r)(n + 2^r)$? As r decreases, the factor b/r increases, but as r increases so does 2^r . The answer depends on whether $b < \lfloor \lg n \rfloor$. If $b < \lfloor \lg n \rfloor$, then $r \leq b$ implies $(n + 2^r) = \Theta(n)$. Thus, choosing $r = b$ yields a running time of $(b/b)(n + 2^b) = \Theta(n)$, which is asymptotically optimal. If $b \geq \lfloor \lg n \rfloor$, then choosing $r = \lfloor \lg n \rfloor$ gives the best running time to within a constant factor, which we can see as follows.¹ Choosing $r = \lfloor \lg n \rfloor$ yields a running time of $\Theta(bn/\lg n)$. As r increases above $\lfloor \lg n \rfloor$, the 2^r term in the numerator increases faster than the r term in the denominator, and so increasing r above $\lfloor \lg n \rfloor$ yields a running time of $\Omega(bn/\lg n)$. If instead r were to decrease below $\lfloor \lg n \rfloor$, then the b/r term increases and the $n + 2^r$ term remains at $\Theta(n)$.

Is radix sort preferable to a comparison-based sorting algorithm, such as quicksort? If $b = O(\lg n)$, as is often the case, and $r \approx \lg n$, then radix sort's running time is $\Theta(n)$, which appears to be better than quicksort's expected running time of $\Theta(n \lg n)$. The constant factors hidden in the Θ -notation differ, however. Although radix sort may make fewer passes than quicksort over the n keys, each pass of radix sort may take significantly longer. Which sorting algorithm to prefer depends on the characteristics of the implementations, of the underlying machine (e.g., quicksort often uses hardware caches more effectively than radix sort), and of the input data. Moreover, the version of radix sort that uses counting sort as the intermediate stable sort does not sort in place, which many of the $\Theta(n \lg n)$ -time comparison sorts do. Thus, when primary memory storage is at a premium, an in-place algorithm such as quicksort could be the better choice.

Exercises

8.3-1

Using Figure 8.3 as a model, illustrate the operation of RADIX-SORT on the following list of English words: COW, DOG, SEA, RUG, ROW, MOB, BOX, TAB, BAR, EAR, TAR, DIG, BIG, TEA, NOW, FOX.

¹ The choice of $r = \lfloor \lg n \rfloor$ assumes that $n > 1$. If $n \leq 1$, there is nothing to sort.

8.3-2

Which of the following sorting algorithms are stable: insertion sort, merge sort, heapsort, and quicksort? Give a simple scheme that makes any comparison sort stable. How much additional time and space does your scheme entail?

8.3-3

Use induction to prove that radix sort works. Where does your proof need the assumption that the intermediate sort is stable?

8.3-4

Suppose that COUNTING-SORT is used as the stable sort within RADIX-SORT. If RADIX-SORT calls COUNTING-SORT d times, then since each call of COUNTING-SORT makes two passes over the data (lines 4–5 and 11–13), altogether $2d$ passes over the data occur. Describe how to reduce the total number of passes to $d + 1$.

8.3-5

Show how to sort n integers in the range 0 to $n^3 - 1$ in $O(n)$ time.

★ 8.3-6

In the first card-sorting algorithm in this section, which sorts on the most significant digit first, exactly how many sorting passes are needed to sort d -digit decimal numbers in the worst case? How many piles of cards does an operator need to keep track of in the worst case?

8.4 Bucket sort

Bucket sort assumes that the input is drawn from a uniform distribution and has an average-case running time of $O(n)$. Like counting sort, bucket sort is fast because it assumes something about the input. Whereas counting sort assumes that the input consists of integers in a small range, bucket sort assumes that the input is generated by a random process that distributes elements uniformly and independently over the interval $[0, 1)$. (See Section C.2 for a definition of a uniform distribution.)

Bucket sort divides the interval $[0, 1)$ into n equal-sized subintervals, or *buckets*, and then distributes the n input numbers into the buckets. Since the inputs are uniformly and independently distributed over $[0, 1)$, we do not expect many numbers to fall into each bucket. To produce the output, we simply sort the numbers in each bucket and then go through the buckets in order, listing the elements in each.

The BUCKET-SORT procedure on the next page assumes that the input is an array $A[1 : n]$ and that each element $A[i]$ in the array satisfies $0 \leq A[i] < 1$. The code requires an auxiliary array $B[0 : n - 1]$ of linked lists (buckets) and assumes

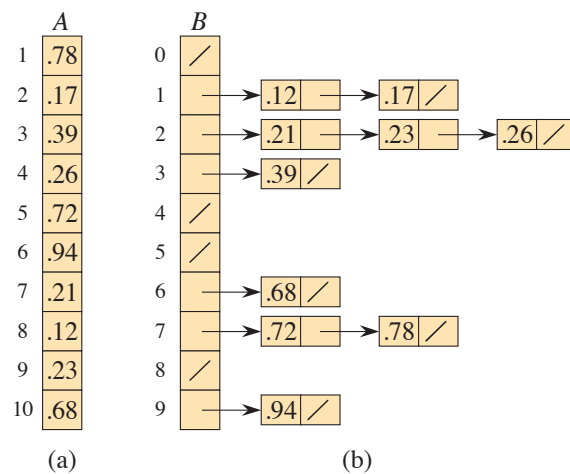


Figure 8.4 The operation of BUCKET-SORT for $n = 10$. (a) The input array $A[1:10]$. (b) The array $B[0:9]$ of sorted lists (buckets) after line 7 of the algorithm, with slashes indicating the end of each bucket. Bucket i holds values in the half-open interval $[i/10, (i+1)/10)$. The sorted output consists of a concatenation of the lists $B[0], B[1], \dots, B[9]$ in order.

that there is a mechanism for maintaining such lists. (Section 10.2 describes how to implement basic operations on linked lists.) Figure 8.4 shows the operation of bucket sort on an input array of 10 numbers.

BUCKET-SORT(A, n)

```

1  let  $B[0:n-1]$  be a new array
2  for  $i = 0$  to  $n-1$ 
3      make  $B[i]$  an empty list
4  for  $i = 1$  to  $n$ 
5      insert  $A[i]$  into list  $B[\lfloor n \cdot A[i] \rfloor]$ 
6  for  $i = 0$  to  $n-1$ 
7      sort list  $B[i]$  with insertion sort
8  concatenate the lists  $B[0], B[1], \dots, B[n-1]$  together in order
9  return the concatenated lists

```

To see that this algorithm works, consider two elements $A[i]$ and $A[j]$. Assume without loss of generality that $A[i] \leq A[j]$. Since $\lfloor n \cdot A[i] \rfloor \leq \lfloor n \cdot A[j] \rfloor$, either element $A[i]$ goes into the same bucket as $A[j]$ or it goes into a bucket with a lower index. If $A[i]$ and $A[j]$ go into the same bucket, then the **for** loop of lines 6–7 puts them into the proper order. If $A[i]$ and $A[j]$ go into different buckets, then line 8 puts them into the proper order. Therefore, bucket sort works correctly.

To analyze the running time, observe that, together, all lines except line 7 take $O(n)$ time in the worst case. We need to analyze the total time taken by the n calls to insertion sort in line 7.

To analyze the cost of the calls to insertion sort, let n_i be the random variable denoting the number of elements placed in bucket $B[i]$. Since insertion sort runs in quadratic time (see Section 2.2), the running time of bucket sort is

$$T(n) = \Theta(n) + \sum_{i=0}^{n-1} O(n_i^2) . \quad (8.1)$$

We now analyze the average-case running time of bucket sort, by computing the expected value of the running time, where we take the expectation over the input distribution. Taking expectations of both sides and using linearity of expectation (equation (C.24) on page 1192), we have

$$\begin{aligned} E[T(n)] &= E \left[\Theta(n) + \sum_{i=0}^{n-1} O(n_i^2) \right] \\ &= \Theta(n) + \sum_{i=0}^{n-1} E[O(n_i^2)] \quad (\text{by linearity of expectation}) \\ &= \Theta(n) + \sum_{i=0}^{n-1} O(E[n_i^2]) \quad (\text{by equation (C.25) on page 1193}) . \end{aligned} \quad (8.2)$$

We claim that

$$E[n_i^2] = 2 - 1/n \quad (8.3)$$

for $i = 0, 1, \dots, n-1$. It is no surprise that each bucket i has the same value of $E[n_i^2]$, since each value in the input array A is equally likely to fall in any bucket.

To prove equation (8.3), view each random variable n_i as the number of successes in n Bernoulli trials (see Section C.4). Success in a trial occurs when an element goes into bucket $B[i]$, with a probability $p = 1/n$ of success and $q = 1 - 1/n$ of failure. A binomial distribution counts n_i , the number of successes, in the n trials. By equations (C.41) and (C.44) on pages 1199–1200, we have $E[n_i] = np = n(1/n) = 1$ and $\text{Var}[n_i] = npq = 1 - 1/n$. Equation (C.31) on page 1194 gives

$$\begin{aligned} E[n_i^2] &= \text{Var}[n_i] + E^2[n_i] \\ &= (1 - 1/n) + 1^2 \\ &= 2 - 1/n , \end{aligned}$$

which proves equation (8.3). Using this expected value in equation (8.2), we get that the average-case running time for bucket sort is $\Theta(n) + n \cdot O(2 - 1/n) = \Theta(n)$.

Even if the input is not drawn from a uniform distribution, bucket sort may still run in linear time. As long as the input has the property that the sum of the squares of the bucket sizes is linear in the total number of elements, equation (8.1) tells us that bucket sort runs in linear time.

Exercises

8.4-1

Using Figure 8.4 as a model, illustrate the operation of BUCKET-SORT on the array $A = \langle .79, .13, .16, .64, .39, .20, .89, .53, .71, .42 \rangle$.

8.4-2

Explain why the worst-case running time for bucket sort is $\Theta(n^2)$. What simple change to the algorithm preserves its linear average-case running time and makes its worst-case running time $O(n \lg n)$?

8.4-3

Let X be a random variable that is equal to the number of heads in two flips of a fair coin. What is $E[X^2]$? What is $E^2[X]$?

8.4-4

An array A of size $n > 10$ is filled in the following way. For each element $A[i]$, choose two random variables x_i and y_i uniformly and independently from $[0, 1)$. Then set

$$A[i] = \frac{\lfloor 10x_i \rfloor}{10} + \frac{y_i}{n}.$$

Modify bucket sort so that it sorts the array A in $O(n)$ expected time.

★ 8.4-5

You are given n points in the unit disk, $p_i = (x_i, y_i)$, such that $0 < x_i^2 + y_i^2 \leq 1$ for $i = 1, 2, \dots, n$. Suppose that the points are uniformly distributed, that is, the probability of finding a point in any region of the disk is proportional to the area of that region. Design an algorithm with an average-case running time of $\Theta(n)$ to sort the n points by their distances $d_i = \sqrt{x_i^2 + y_i^2}$ from the origin. (*Hint:* Design the bucket sizes in BUCKET-SORT to reflect the uniform distribution of the points in the unit disk.)

★ 8.4-6

A *probability distribution function* $P(x)$ for a random variable X is defined by $P(x) = \Pr\{X \leq x\}$. Suppose that you draw a list of n random variables